

The iterated expansion for a general base: LogBase and the weighted primitives F_γ

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Abstract

Unit 43's complete polynomial-in-log expansion of the iterated primitives $H_{m+1}(x) = \int_0^x H_m(t)/t dt$ used only four properties of the base H_0 . This unit isolates them as a predicate **LogBase** **G A M C** (measurable, $0 \leq G \leq A$, $G(x) \leq Cx^\delta$ on $(0,1]$ with $\delta > 0$, $|G(L) - A| \leq M/L$ for $L \geq 1$), redoes the construction with all integrals over $(0, x]$, and proves $|H_m^G(L) - \sum_{i \leq m} c_{m,i} \log^i L| \leq M/L$ with the same recursive coefficients and $c_{m,m} = A/m!$. The weighted primitive F_γ ($\gamma > -1$) is an admissible base with $(A, M, C, \delta) = (A_\gamma, A_{\gamma+1}, E/(\gamma+1), \gamma+1)$, which is what the d -fold general- (k, h) theory needs. Zero **sorry/axiom**.

1 Setting

The transfer step $\theta_{m+1}(L) = -\int_L^\infty \theta_m/x$ requires, at each level, that H_m/t be integrable on $(0, L]$ and that H_m be measurable. With no continuity available for a general base, both are obtained structurally: H_{m+1} is monotone (integral of a nonnegative function over a growing set), hence measurable and bounded on $(1, L]$ by $H_{m+1}(L)$, while near 0 the bound $H_m \leq (C/\delta^m)x^\delta$ propagates because $\int_0^x t^{\delta-1} = x^\delta/\delta$.

2 The things formalised

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noncomputable def iterDivPrim (G :  $\mathbb{R} \rightarrow \mathbb{R}$ ) :  $\mathbb{R} \rightarrow \mathbb{R}$ 
| 0 => G
| m + 1 => fun x =>  $\int_L^\infty \theta_m/x$  t in Ioc 0 x, iterDivPrim G m t / t
structure LogBase (G :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (A M C :  $\mathbb{R}$ ) : Prop where
  measurable : Measurable G
  nonneg :  $\forall x, 0 \leq G x$ 
  le_mass :  $\forall x, G x \leq A$ 
  le_pow :  $\forall x, 0 < x \rightarrow x^{-1} \rightarrow G x \leq C * x^\delta$ 
  _pos : 0 <
  tail :  $\forall L, 1 \leq L \rightarrow |G L - A| \leq M / L$ 
theorem iterDivPrim_props (hG : LogBase G A M C) (m :  $\mathbb{N}$ ) :
  Measurable (iterDivPrim G m) (x, 0 < x) (iterDivPrim G m x)
  (x, 0 < x  $\rightarrow x^{-1} \rightarrow$  iterDivPrim G m x  $\leq C / \delta^m * x^\delta$ )
  (L, IntegrableOn (fun t => iterDivPrim G m t / t) (Ioc 0 L))
theorem iterDivPrim_expansion (hG : LogBase G A M C) (m :  $\mathbb{N}$ ) (L :  $\mathbb{R}$ ) (hL : 1  $\leq$  L) :
  |iterDivPrim G m L -  $\sum_{i \leq m} c_{m,i} \log^i L| \leq M / L$ 
  Finset.range (m + 1), divCoeff G A m i * Real.log L ^ i |
theorem weightedPrimitive_logBase (a :  $\mathbb{R}$ ) (h : 0 <  $\delta$ ) (h : -1 <  $\gamma$ ) :
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LogBase (weightedPrimitive a) (weightedMass a) (weightedMass a ( + 1))
  (Real.exp ( * a ^ 2 / 2) / ( + 1)) ( + 1)
```

3 Proof strategy

A single helper, `div_integrableOn_of_bounds`, turns “measurable, nonnegative, $\leq C't^\delta$ near 0, bounded on $(1, L]$ ” into integrability of H/t on $(0, L]$. The per-level induction `iterDivPrim_props` threads exactly these four facts, using monotonicity for measurability and for the bound on $(1, L]$. The coefficient recursion, remainder recursion and uniform bound are then unit 43’s proofs with `iterLogPrim` replaced by `iterDivPrim` (no interval-integral conversion is needed since the definition is already over `Ioc`) and with continuity replaced by the measurability from the props. The base instance for F_γ collects unit 45’s bounds; for $x < 0$ the primitive is the integral over an empty set.

4 Roadblocks and resolutions

- `le_or_lt` no longer exists on this pin; use `le_or_gt`.
- `simpa` could not see through `iterDivPrim G 0`; rewrite with the definitional lemma first.
- An unused `LogBase` hypothesis in the tail bound was flagged by the linter; the lemma is stated for arbitrary G, A, M .

5 What was Mathlib, what was new

Mathlib: `Monotone.measurable`, `setIntegral_mono_set`, `div_le_self`, `Real.rpow_add_one`, `integral_rpow`, `Finset.measurable_sum`, `Real.measurable_log`. New: the `LogBase` abstraction, the per-level invariant, and the base-independent expansion.

6 Lessons

Abstracting after two concrete instances (units 43 and 45) is the right moment: the hypotheses were already known to be sufficient, so the abstraction cost one file and no new mathematics. Monotonicity is again the key to measurability and boundedness without continuity.

7 Outcome

The complete expansion of the iterated primitives is available for every admissible base, in particular for every weighted primitive F_γ . Next: the d -fold chart recursion with general (k, h) and all candidate exponents equal, reducing to `iterDivPrim (weightedPrimitive a (p-1))` and giving $Z_d(n) \sim (\prod_i k_i^{-1}) \frac{A_{p-1}}{2^{d-1}(d-1)!} n^{-p/2} (\log n)^{d-1}$.